# ZHOU, CONGHAO

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## CONTACT

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## RESEARCH INTERESTS

dark matter · gravitational lensing · galaxy clusters · large scale structure
shear calibration · image simulation

## EDUCATION

### University of California, Santa Cruz

*Expected 2027* Ph.D. in Physics
Thesis: *No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*
Advisors: Tesla E. Jeltema, Alexie S. Leauthaud

### Duke University

*2020* B.A. in Physics with Highest Distinction
Thesis: *Intrinsic Alignment of redMaPPer Clusters in the Dark Energy Survey*
Advisor: Michael A. Troxel

## SCIENCE PUBLICATIONS WITH MAJOR CONTRIBUTIONS

† STUDENT ADVISED OR CO-ADVISED

1. The Stellar-Mass Dependence of Feedback-Driven Outflows in DESI DR1 Star-Forming Galaxies
   arXiv:2611.02580 (2026), *submitted to Astrophys. J. Lett.*
   Y. Tang, Y. Luo, **C. Zhou**
2. Cluster finding with outskirt stellar masses and percolation
   arXiv:2608.19768 (2026), *submitted to Open J. Astrophys.*
   P. Avalos†, **C. Zhou**, T. Jeltema, K. Leidig, S. Xu et al.
3. Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal
   arXiv:2606.21141 (2026), *submitted to Astrophys. J.*
   **C. Zhou**, X. Li et al.
4. Relationship between 2D and 3D galaxy stellar mass and correlations with halo mass
   J. Cosmol. Astropart. Phys. 10, 014 (2025) [arXiv:2502.05158]
   **C. Zhou**, A. Leauthaud, S. Xu, B. Diemer, S. Huang et al.
5. Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data
   Phys. Rev. D 110, 103508 (2024) [arXiv:2312.11789], *Editors' Suggestion*
   **C. Zhou**, H.-Y. Wu, A.N. Salcedo, S. Grandis, T. Jeltema, A. Leauthaud et al.
6. The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters
   Mon. Not. R. Astron. Soc. 526, 323 (2023) [arXiv:2302.12325]
   **C. Zhou**, A. Tong, M.A. Troxel, J. Blazek, C. Lin et al.

## TECHNICAL PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Surface brightness profiles around massive galaxies from LSSTCam coadds
   LSST Technote SOTN-007 (2026)
   **C. Zhou**, T. Jeltema, A. von der Linden, L.S. Kelvin et al.

2. Surface brightness profiles around massive galaxies in LSSTComCam data
   LSST Technote SITCOMTN-165 (2025)
   **C. Zhou**, T. Jeltema, A. von der Linden, L.S. Kelvin et al.

## CONTRIBUTING AUTHOR SCIENCE PUBLICATIONS

1. The Merian Survey: A Statistical Census of Bright Satellites of Sub-Milky Way-Mass Galaxies
   arXiv:2610.03981 (2026)
   L.S. Mandacarú Guerra et al.

2. Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics
   Astrophys. J. 1009, 225 (2026) [arXiv:2510.17996]
   Y. Luo et al.

3. The Connection between Halo Assembly History and the Stellar Distribution I: The Central Stellar Mass–Richness Plane
   arXiv:2609.14488 (2026)
   S. Xu et al.

4. Probing Projection Effects in Optically-Selected Clusters with Velocity Dispersions Outliers
   arXiv:2609.08095 (2026)
   K. Graham et al.

5. The Vera C. Rubin Observatory Data Preview 1
   Astron. J. 171, 360 (2026) [arXiv:2603.23786]
   Vera C. Rubin Observatory Team et al.

6. DESI-DR1 3 × 2-pt analysis: consistent cosmology across weak lensing surveys
   Open J. Astrophys. 9 (2026) [arXiv:2512.15960]
   A. Porredon et al.

7. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)
   Astrophys. J. Lett. 1001, L35 (2026) [arXiv:2507.13409]
   C.O. Chandler et al.

8. Data Release 1 of the Dark Energy Spectroscopic Instrument
   Astron. J. 171, 285 (2026) [arXiv:2503.14745]
   DESI Collaboration et al.

9. Impact of projection-induced optical selection bias on the weak lensing mass calibration of galaxy clusters
   Phys. Rev. D 113, 063559 (2026) [arXiv:2510.00753]
   T.N. Nde et al.

10. Reaching for the Edge II: Stellar Halos out to Large Radii as a Tracer of Dark Matter Halo Mass
    arXiv:2511.10723 (2025)
    K. Leidig et al.

11. Cosmological Constraints from Dark Energy Survey Year 1 Cluster Lensing and Abundances with Simulation-based Forward-Modeling
    arXiv:2510.25706 (2025)
    A.N. Salcedo et al.

12. Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering
    Phys. Rev. D 112, 083535 (2025) [arXiv:2503.13632]
    T.M.C. Abbott et al.

13. The Outskirt Stellar Mass of Low-redshift Massive Galaxies Is an Excellent Halo Mass Proxy in Illustris/IllustrisTNG Simulations
    Astrophys. J. 990, 127 (2025) [arXiv:2412.03406]
    S. Xu et al.

14. Association between optically identified galaxy clusters and the underlying dark matter halos
Phys. Rev. D 112, 043517 (2025) [arXiv:2506.17526]
S. Cao et al.

15. Improving galaxy cluster selection with the outskirt stellar mass of galaxies
Phys. Rev. D 111, 123524 (2025) [arXiv:2410.20205]
M. Kwiecien et al.

16. Optical galaxy cluster mock catalogs with realistic projection effects: Validations with the SDSS clusters
Phys. Rev. D 111, 063502 (2025) [arXiv:2410.02497]
A. Lee et al.

17. Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations
Phys. Rev. D 109, 063534 (2024) [arXiv:2310.17557]
M. Gatti et al.

18. Dark Energy Survey Year 3 Results: Deep Field optical + near-infrared images and catalogue
Mon. Not. R. Astron. Soc. 509, 3547 (2022) [arXiv:2012.12824]
W.G. Hartley et al.

## CONTRIBUTING AUTHOR METHODOLOGY PUBLICATIONS

1. Getting Ready For Analysing Rubin Data: CLMM v2, The Updated LSST-DESC Cluster Weak Lensing Mass Modeling Library For Cosmology
arXiv:2610.37455 (2026)
M. Aguena, C. Combet, S. Fu, C. Payerne, M. Ricci et al.

2. OASIS: Observation-Aware Simulation-Based Inference via Distributional Matching
arXiv:2606.22572 (2026)
A. Farahi, **C. Zhou**, R. Vashistha

3. Building an Efficient Cluster Cosmology Software Package for Modeling Cluster Counts and Lensing
arXiv:2309.06593 (2023)
M. Aguena, O. Alves, J. Annis, D. Bacon, S. Bocquet et al.

## COMPETITIVE FUNDING

*2026–2027* **Chancellor's Dissertation Year Fellowship**, UC Santa Cruz, $33,000

*2020* **Rodney I. McCormick Award**, Duke University, $1,500

## TALKS

*nov. 2026* **KIPAC, Stanford University and SLAC**, Tea Talk
*No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*

*oct. 2026* **Kavli IPMU, University of Tokyo**, Astro Seminar
*No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*

*aug. 2026* **Tsinghua University**, Department of Astronomy Seminar
*No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*

*jul. 2026* **Tsung-Dao Lee Institute**, Astronomy Seminar
*No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*

*jul. 2026* **Shanghai Jiao Tong University**, Department of Astronomy Seminar
*No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*

*may 2026* **South Pole Telescope Collaboration**, Cluster Working Group Meeting
*No Cluster Left Behind: Joint Mass Calibration for DES and SPT Clusters*

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| *dec. 2024* | **University of Arizona**, Theoretical Astrophysics and Cosmology Seminar<br>*Cluster Cosmology: Pixel to Object, CMB to Stellar Mass* |
| *nov. 2024* | **University of California, Davis**, Cosmology Seminar<br>*Optical Cluster Cosmology: Selection to IA, Pixel to Object* |
| *apr. 2024* | **CMB-S4 Collaboration**, Cluster AWG Meeting<br>*Calibrating Optical Selection Bias with Multiwavelength Data* |
| *mar. 2024* | **Brown University**, Center for the Fundamental Physics of the Universe Seminar<br>*Cluster Cosmology Beyond Richness* |
| *mar. 2024* | **University of Pennsylvania**, Journal Club<br>*Response Clustering Bias* |
| *oct. 2023* | **Tsinghua University**, Cosmology Seminar<br>*Intrinsic Alignment in DES Y1 redMaPPer Clusters* |
| *oct. 2023* | **Stony Brook University & Brookhaven National Lab**, Cosmology Seminar<br>*Constraining Optical Selection Bias in Cluster Lensing w/ Multiwavelength Data* |

## OBSERVING EXPERIENCE

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| *10.5 effective nights* | Dark Energy Spectroscopic Instrument, Mayall 4m at Kitt Peak |
| *3 nights* | DECaLS Intermediate Band Imaging Survey, Blanco 4m at CTIO |

## SCIENCE COLLABORATIONS

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|---|---|
| *2026–present* | Roman Science Collaboration |
| *2024–present* | Rubin Observatory Commissioning, In-Kind Contributor |
| *2023–present* | LSST Galaxies Science Collaboration |
| *2023–present* | LSST Informatics & Statistics Science Collaboration |
| *2020–present* | LSST Dark Energy Science Collaboration, Full Member since 2024 |
| *2020–present* | Dark Energy Spectroscopic Instrument, Sponsored Member |
| *2018–present* | Dark Energy Survey, Student Member |

## PROFESSIONAL SOCIETIES

| | |
|---|---|
| *2026–present* | Sigma Xi, Member |
| *2024–present* | American Astronomical Society, Student Member |
| *2024–present* | American Physical Society, Student Member |

## OUTREACH

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| *2023–2024* | Starlight Over Street Light |
| *2022* | UC Santa Cruz Physics Outreach Taskforce |
| *2022* | Science Internship Program Mentor |
| *2022* | UC Santa Cruz Physics Mentoring Program Mentor |

## COLLABORATION SERVICE

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| *jun. 2026–present* | LSST DESC Speaker's Bureau |
| *2025, 2026* | LSST DESC Collaboration Council Nomination Committee (Chair, 2026) |
| *jul. 2022–present* | LSST DESC Mentoring Committee |
| *apr.2022–may.2025* | DES Early Career Scientists Committee (Chair, may 2024 – may 2025) |

## SEMINAR & MEETING ORGANIZATION

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| *2025 Winter* | UC Santa Cruz Physics Colloquium Committee |
| *sep.2023–jun.2024* | Organizer, CosmoGal Seminar, UC Santa Cruz |
| *jul.2023–jun.2024* | Organizer, BCG Seminar, Boise State |
| *jan. 2023* | LOC, APS Conference for Undergraduate Women in Physics at UC Santa Cruz |
| *aug. 2022* | SOC, LSST DESC 2022 Chicago Meeting |

## REFEREEING

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| *2024–present* | Astronomy & Astrophysics |

## TEACHING EXPERIENCE

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|---|---|
| *2024 Spring* | Application of Quantum Materials (Grader) |
| *2024 Winter* | Electricity, Magnetism, and Optics (TA) |
| *2022 Fall* | Quantum Mechanics (TA) |
| *2022 Summer* | Intermediate Physics Laboratory (TA) |
| *2022 Spring* | Advanced Astrophysics Laboratory (TA) |
| *3 times* | Introductory Physics Laboratory (TA) |

November 6, 2026